📘 Product Requirements Document

ICET 2026 Mock Preparation Portal

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Table of Contents

1. Product Overview
2. Goals & Success Metrics
3. User Personas
4. Information Architecture & Page Structure
5. Detailed Feature Requirements
    - 5.1 Landing Page
    - 5.2 Test Selection Page
    - 5.3 Test Interface (Exam Simulator)
    - 5.4 Result & Explanation Page
    - 5.5 ICET Guide Page
    - 5.6 Mock Full Tests (Gemini-Generated)
    - 5.7 APICET Previous Year Papers Section
    - 5.8 API Key Configuration
6. ICET 2026 Exam Pattern Reference
7. PDF Content & Naming Convention
8. Gemini AI Integration Spec
9. UI/UX Design System (Quality Theme)
10. Component Library & Interaction Patterns
11. Technical Architecture
12. Data Flow & State Management
13. Performance Requirements
14. Accessibility & Responsiveness
15. Out of Scope

16. Open Questions

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1. Product Overview

1.1 Purpose

A dedicated, full-featured web-based mock examination portal for students preparing for **AP ICET 2026** (Andhra Pradesh Integrated Common Entrance Test). The portal simulates the real exam environment, provides AI-powered answers and explanations via **Google Gemini**, and includes a curated study guide to help students prepare holistically.

1.2 Core Value Proposition

- Authentic exam simulation with timed, section-wise tests
- AI-generated motivational quotes on every visit (Gemini)
- AI-powered answer keys with step-by-step explanations (Gemini)
- 21 shift-wise question papers + 2 full Grand Mock tests
- Complete APICET previous year papers with structured navigation
- Section-filtered question palette (just like the real exam)
- Zero backend required — fully client-side with PDF + Gemini

1.3 Target Audience

Students appearing for AP ICET 2026; coaching institutes; self-study aspirants in Andhra Pradesh and Telangana.

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2. Goals & Success Metrics

Goal	Success Metric
Students complete at least one mock test	⩾ 80% session completion rate
AI explanations load within 3 seconds	Gemini API response < 3s per question
Motivational quote refreshes on every visit	100% new quote per page load/refresh
All 21 PDFs load correctly	0 broken paper links
Mock test timer matches real exam	Timer accurate ±1 second

Goal	Success Metric
Mobile usability	Functional on 360px minimum width

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3. User Personas

Persona 1 — Priya (Self-Study Aspirant)

- Preparing from home, limited coaching access
- Needs: Clear guidance, timed practice, Telugu-free papers
- Goal: Score 90+ to get into a top MBA college

Persona 2 — Ravi (Last-Minute Crammer)

- 2 weeks before exam, wants quick mock tests
- Needs: Full-length timed tests, instant answer keys
- Goal: Identify weak areas quickly

Persona 3 — Coaching Instructor

- Wants to share this portal with 30+ students
- Needs: Reliable UI, consistent content across sessions
- Goal: Replace printed papers with digital practice

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4. Information Architecture & Page Structure

```
ICET Mock Prep Portal
|
├── [PAGE 1] Landing Page
|   ├── Hero Text — "ICET 2026 Preparation Hub" + Tagline
|   ├── AI Motivational Quote (Gemini, refreshes each visit)
|   ├── Gemini API Key Input (first-time / settings)
|   └── CTA Button — "Begin Your Journey →"
|
├── [PAGE 2] Test Selection Hub
|   ├── Navigation Tabs: [Shift-wise Papers | Grand Mocks | Previous Year | Guide]
|   ├── Section: 21 Shift-wise Papers (PDF-named buttons)
|   ├── Section: 2 Gemini Grand Mock Tests
```

```
|   ├── Section: APICET Previous Year Papers
|   |   ├── Set 1 (Years split)
|   |   └── Set 2 (Years split)
|   └── Section: ICET Guide
|
├── [PAGE 3] Exam Simulator Interface
|   ├── Header Bar — Section Dropdown + Timer
|   ├── Left Panel — Question Palette (numbered, color-coded)
|   ├── Right Panel — Question + 4 Options
|   └── Footer Bar — Prev | Clear | Submit | Review | Next
|
├── [PAGE 4] Results & Explanation Page
|   ├── Score Card (Section-wise breakdown)
|   ├── Question-by-Question Review
|   |   ├── Your Answer (highlighted)
|   |   ├── Correct Answer (highlighted)
|   |   └── AI Explanation (Gemini-generated)
|   └── Actions: Retry | Back to Hub
|
└── [PAGE 5] ICET Guide
    ├── Exam Pattern Overview
    ├── Section-wise Strategy
    ├── Important Topics Checklist
    ├── Time Management Tips
    └── PDF Resource Downloads
```

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5. Detailed Feature Requirements

5.1 Landing Page

5.1.1 Hero Section

- **Primary Heading:** `"ICET 2026 — Your Gateway to MBA Success"` (large, animated entrance)
- **Subtitle:** `"Master Every Shift. Crack Every Section."` (fade in, 300ms delay)
- **Visual:** Subtle animated geometric/academic background — no stock photos

5.1.2 Motivational Quote Block

- **Source:** Google Gemini API (`gemini-2.0-flash` or latest)

- **Prompt to Gemini:** `"Generate a unique, powerful motivational quote (max 2 sentences) specifically for a student preparing for a competitive MBA entrance exam like ICET. Be creative, not generic. Return ONLY the quote and author attribution."`
- **Behavior:** Generates a fresh quote on every page load and every manual refresh
- **Display:** Stylized blockquote component with quotation marks, name attribution below
- **Fallback:** If API fails, rotate from a hardcoded list of 20 quotes

5.1.3 API Key Input

- **When to show:** On first visit (no key stored) OR via "Settings" icon
- **UI:** Clean modal overlay — text input labeled `"Enter your Gemini API Key"`, save button, close
- **Storage:** `localStorage` key: `icet_gemini_api_key`
- **Validation:** Attempt a lightweight test call; show green ✓ or red ✗ status
- **Note:** Key is never sent anywhere except Gemini's own API

5.1.4 CTA Button

- **Label:** `"Begin Your Journey →"` or `"Enter the Arena →"` (visually bold)
- **Behavior:** Navigate to Page 2 (Test Selection Hub)
- **Design:** Prominent, animated hover state, NOT generic flat button

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5.2 Test Selection Page

5.2.1 Page Header

- Compact top bar: Portal logo/name + Settings icon (for API key) + optional user name display

5.2.2 Navigation Tabs

Four clearly labeled tabs:

Tab Label	Content
📄 Shift Papers	21 shift-wise PDF-based tests
🏆 Grand Mocks	2 AI-generated full-length mock tests
📚 Previous Year	APICET solved papers (Set 1 & Set 2)
📖 Study Guide	ICET 2026 preparation guide

5.2.3 Shift Papers Grid (21 Papers)

- **Layout:** Responsive grid — 3 columns desktop, 2 tablet, 1 mobile
- **Button design:** Card-style button, showing:
  - Paper number (e.g., Paper 01)
  - PDF file name as title (e.g., `ICET_Shift1_2024_Morning`)
  - Pill badge showing section count or question count if extractable
  - Status icon: Not Attempted (grey) / Attempted (green tick) / In Progress (orange)
- **Naming:** Button title = exact PDF filename (without `.pdf` extension), formatted with spaces
- **On click:** Start exam in the Exam Simulator (Page 3)

5.2.4 Grand Mock Tests Section

- **2 Tests:** `Grand Mock Test 1` and `Grand Mock Test 2`
- **Source:** AI-generated by Gemini from content across all 21 PDFs
- **Pattern:** Follows official ICET 2026 pattern exactly (200 questions, 150 min)
- **Badge:** "AI Generated" label clearly visible
- **Status tracking:** Same as shift papers

5.2.5 APICET Previous Year Papers

- **Division:** Because the PDF is large, split into two navigable sets:
  - **Set 1:** Earlier years (to be defined when PDF is provided)
  - **Set 2:** Recent years (to be defined when PDF is provided)
- **Structure:** Accordion or sub-tab per year with question count shown
- **Each year:** Opens in the Exam Simulator with the same interface

5.2.6 Progress Overview Panel

- Top of selection page shows a mini-dashboard:
  - Papers attempted / 21
  - Average score %
  - Best section performance
  - Streak: Days practiced

5.3 Exam Simulator Interface

This is the core screen. Based on the reference image provided.

5.3.1 Layout Structure

```
┌──────────────────────────────────────────────────────────────────────┐
│  [Section Dropdown ▼]      Test Name          ⏱ HH:MM:SS  [⚙]     │
├──────────────────┬───────────────────────────────────────────────────┤
│                  │                                                   │
│  QUESTION        │   Q.No. X — Question Text                        │
│  PALETTE         │                                                   │
│                  │   ○ Option A                                      │
│  [ 1 ][ 2 ][ 3 ] │   ○ Option B                                      │
│  [ 4 ][ 5 ][ 6 ] │   ○ Option C                                      │
│  ...             │   ○ Option D                                      │
│                  │                                                   │
│  Legend:         │                                                   │
│  □ Not Visited   │                                                   │
│  🔴 Not Answered │                                                   │
│  🟢 Answered     │                                                   │
│  🔵 Ans+Review   │                                                   │
│  🟡 Marked       │                                                   │
├──────────────────┴───────────────────────────────────────────────────┤
│  [◀ Prev]  [Clear]          [Submit]        [Review ⭐]  [Next ▶]  │
└──────────────────────────────────────────────────────────────────────┘
```

5.3.2 Header Bar Components

Component	Spec
Section Dropdown	Filters question palette by section (Sec A / Sec B / Sec C / All). Matches ICET section structure
Test Name	Centered — PDF name or Mock test name
Timer	Counts DOWN from 90:00 for shift papers / 150:00 for grand mocks. Format HH:MM:SS. Turns orange at 10 min, red at 5 min
Settings icon	Pause test, change API key

5.3.3 Left Panel — Question Palette

- Scrollable grid of numbered boxes (up to 200 questions)
- **Color Coding (matches real ICET):**
  - ⬜ **White / Grey outline** = Not Visited
  - 🔴 **Red** = Visited but Not Answered
  - 🟢 **Green** = Answered
  - 🔵 **Blue/Purple** = Answered and Marked for Review
  - 🟡 **Yellow/Orange** = Marked for Review (without answer)
- Clicking a number jumps directly to that question
- Section filter from dropdown updates palette to highlight only that section's questions
- Bottom legend explaining all colors

5.3.4 Right Panel — Question Area

- **Question number** shown as `Q. 1 of 200`
- **Question text** — clean, readable, large font (min 16px)
- **4 Radio Options** (A, B, C, D) — each is a tappable full-width row (not just the radio circle)
- Options have hover highlight, selected state shows filled circle + highlighted background
- Long questions scroll within the panel without affecting palette

5.3.5 Footer Navigation Bar

Button	Label	Color	Behavior
Previous	`◀ Prev`	Muted/Amber	Go to previous question
Clear	`Clear`	Grey	Deselect current answer
Submit	`Submit`	Green	Confirm submission modal
Review	`Mark Review ⭐`	Blue	Flag question for review
Next	`Next ▶`	Orange/Teal	Go to next question

5.3.6 Submit Confirmation Modal

Before final submission:

- Summary: Answered X / Total Y
- Not answered: Z
- Marked for review: W
- Two buttons: `Go Back` and `Submit Final`

5.3.7 Auto-Submit

When timer reaches 00:00:00, auto-submit fires with a toast notification: `"Time's up! Submitting your paper..."` and transitions to Results page.

5.4 Result & Explanation Page

5.4.1 Score Card Section

- **Overall Score:** Large number display (e.g., `142 / 200`)
- **Section Breakdown Table:**

Section	Questions	Attempted	Correct	Wrong	Score
Section A: Analytical Ability	75	—	—	—	—
Section B: Mathematical Ability	75	—	—	—	—
Section C: Communication Ability	50	—	—	—	—
Total	**200**	—	—	—	—

- **Visual:** Donut chart or bar chart of section performance
- **Time Taken** vs **Total Time** displayed

5.4.2 Question-by-Question Review

For each question, display:

- Question text
- Your selected answer (highlighted in red if wrong, green if correct)
- Correct answer (highlighted in green)
- **AI Explanation block** — "Powered by Gemini" badge
  - Step-by-step reasoning
  - Formula or concept reference (for Math sections)
  - Load on demand (lazy load with "Show Explanation" toggle to save API calls)

5.4.3 Filter Controls on Review Page

- Show: All / Correct / Wrong / Skipped / Marked
- Section filter

5.5 ICET Guide Page

A comprehensive study reference, always accessible from the navigation tab.

5.5.1 Sections in the Guide

A. Exam Pattern Overview

- Full ICET 2026 exam structure table (see Section 6 of this PRD)
- Scoring: +1 for correct, 0 for incorrect (no negative marking)

- Total: 200 marks, 150 minutes

B. Section-wise Strategy Cards

- Section A (Analytical): Data Sufficiency, Sequences, Date/Time puzzles
- Section B (Mathematical): Arithmetic, Algebra, Geometry, Statistics
- Section C (Communication): Vocabulary, Grammar, Reading Comprehension, Business Terminology

C. Important Topics Checklist

- Interactive checklist students can tick off
- Persists in localStorage

D. Time Management Tips

- Recommended time per section
- Question attempt strategy
- Review flagging strategy

E. PDF Resource Downloads

- Direct download links to all 21 PDFs (as uploaded by user)
- APICET Previous Year Papers (Set 1 & Set 2)
- Business & Computer Terminology PDF

5.6 Mock Full Tests (Gemini-Generated)

5.6.1 Generation Logic

- **Trigger:** User clicks "Grand Mock Test 1" or "Grand Mock Test 2"
- **Pre-generation check:** If already generated and cached in localStorage, load cached version
- **Gemini Prompt Structure:**

```
You are generating a full-length ICET 2026 mock test.
Exam Pattern: 200 questions — Section A (75 Qs, Analytical Ability),
Section B (75 Qs, Mathematical Ability), Section C (50 Qs, Communication Ability).
Generate [section] questions.
Each question must have:
- question_text (English only)
- 4 options (A, B, C, D)
- correct_answer (A/B/C/D)
- explanation (step-by-step, 3–5 sentences)
- section (A/B/C)
- topic (e.g., "Arithmetic", "Grammar")
Return ONLY valid JSON array. No markdown, no preamble.
```

- **Split into 3 API calls** (one per section) to avoid token limits
- **Business & Computer Terminology questions** (Section C): Alternating vocabulary → definition format and definition → term format

5.6.2 Generation UX

- Loading screen with animated progress: "Generating Section A... (1/3)"
- Total generation time indicator
- Cache result in localStorage under key `mock_test_1` / `mock_test_2`
- Option to "Regenerate" (clears cache)

5.7 APICET Previous Year Papers

5.7.1 Structure

Because the PDF is large, divide into two sets on the selection page:

Set 1 — Earlier Years

- Sub-buttons per year available in the PDF
- Each year: labeled `"APICET YYYY — Morning Shift"` etc.

Set 2 — Recent Years

- Same structure as Set 1

5.7.2 Display

- Accordion layout per year: click to expand and see available shifts/sets
- Question count shown per test
- "Solved" badge with AI explanations available

5.8 API Key Configuration

Aspect	Spec
Input location	Landing page (first visit modal) + Settings icon on all pages
Storage	`localStorage` — key: `icet_gemini_api_key`
Visibility toggle	Show/hide password field
Validation	Make a minimal test call to Gemini
Error handling	Clear message if key invalid: "Invalid API key. Please check and try again."
Privacy notice	Display: "Your API key is stored only on your device and sent only to Google Gemini."

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6. ICET 2026 Exam Pattern Reference

Section	Name	Questions	Marks	Time (Recommended)
Section A	Analytical Ability	75	75	50 min
— Part 1	Data Sufficiency	20	20	
— Part 2	Problem Solving (Sequences, Coding, Date-Time, etc.)	55	55	
Section B	Mathematical Ability	75	75	60 min
— Part 1	Arithmetical Ability	35	35	
— Part 2	Algebraical and Geometrical Ability	30	30	

Section	Name	Questions	Marks	Time (Recommended)
— Part 3	Statistical Ability	10	10	
Section C	Communication Ability	50	50	40 min
— Part 1	Vocabulary	10	10	
— Part 2	Business & Computer Terminology	10	10	
— Part 3	Functional Grammar	15	15	
— Part 4	Reading Comprehension	15	15	
TOTAL		**200**	**200**	**150 min**

Scoring: +1 correct, 0 for wrong/unattempted (no negative marking)

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7. PDF Content & Naming Convention

7.1 The 21 Shift-wise Papers

- User will supply 21 PDFs
- Button title = filename without `.pdf`, spaces replacing underscores
- Example: `ICET_2024_Shift1_FN.pdf` → Button: `ICET 2024 Shift1 FN`

7.2 Telugu Language Filter

- When extracting questions from PDFs:
  - **Remove** all Telugu-script lines, paragraphs, or options
  - **Keep** only English-language question text and options
  - If a question has no English option text at all → skip it
  - If only some options are Telugu → keep only English options, mark the Telugu ones as unavailable

7.3 Business & Computer Terminology PDF

- Question generation format alternates:
  - **Odd questions:** "What is the meaning of [TERM]?" → 4 definition options

    - **Even questions:** "Which term best describes [DEFINITION]?" → 4 term options
- Source: This specific PDF content only

7.4 APICET Previous Year Solved Papers

- PDF is large — divide by year into logical groupings
- Set 1 and Set 2 division point to be confirmed once PDF is provided
- Each year's paper has: Shift indicator, full solutions, AI explanation layer

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8. Gemini AI Integration Spec

8.1 Model

- Use: `gemini-2.0-flash` (fast, cost-effective, sufficient quality)
- Fallback: `gemini-1.5-flash`

8.2 API Calls Summary

Feature	Prompt type	Frequency
Motivational Quote	Single-turn	Every page load
Mock Test Generation (per section)	Single-turn, JSON output	Once per mock (cached)
Per-question Explanation	Single-turn, JSON	On-demand per question
Business Terminology Questions	Single-turn, JSON array	Once (cached)

8.3 Response Format for Questions (JSON Schema)

```
json
```

```
[
  {
    "id": 1,
    "section": "B",
    "topic": "Arithmetic",
    "question_text": "What is 15% of 360?",
    "options": {
      "A": "48",
      "B": "54",
      "C": "56",
      "D": "60"
    },
    "correct_answer": "B",
    "explanation": "15% of 360 = (15/100) × 360 = 54. Therefore the answer is B."
  }
]
```

8.4 Error Handling

- API key missing → redirect to key input modal
- Rate limit (429) → show toast: "Generating too fast. Retrying in 5 seconds..."
- Network failure → fallback to cached content if available
- Malformed JSON response → retry once; if fails again, show error with retry button

8.5 Caching Strategy

- Mock tests stored in `localStorage` under `icet_mock_1`, `icet_mock_2`
- Explanations cached individually: `explanation_[paperName]_[questionId]`
- Quote cached per session only (not persisted)

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9. UI/UX Design System (Quality Theme)

9.1 Design Philosophy

Theme Name: "Academic Excellence"
Aesthetic Direction: Clean academic + modern digital — inspired by top-tier exam portals (GATE, CAT online). Professional, confidence-inspiring, distraction-free. NOT flashy or game-like.

9.2 Color Palette

Token	Value	Usage
`--color-primary`	`#1A3C6E` (Deep Navy)	Primary actions, headers
`--color-secondary`	`#E8700A` (Warm Amber)	Accent, CTAs, highlights
`--color-success`	`#2E8B57` (Forest Green)	Correct answers, answered questions
`--color-danger`	`#CC3333` (Clear Red)	Wrong answers, not-answered
`--color-review`	`#1565C0` (Royal Blue)	Marked for review
`--color-warn`	`#F4A91C` (Gold)	Timer warning, mark-only
`--color-bg-main`	`#F4F6F9` (Light Slate)	Page background
`--color-bg-card`	`#FFFFFF`	Card / panel backgrounds
`--color-bg-sidebar`	`#EAECF0`	Left palette panel
`--color-text-primary`	`#1A1A2E`	All body text
`--color-text-secondary`	`#555C70`	Labels, captions
`--color-border`	`#D0D7E2`	All borders

9.3 Typography

Role	Font	Size	Weight
Display / Hero	`Playfair Display`	48–64px	700
Section Headings	`Libre Baskerville`	24–32px	600
Body / Questions	`Source Serif Pro`	16–18px	400
UI Labels / Buttons	`DM Sans`	13–15px	500–600
Timer	`JetBrains Mono`	22px	700
Palette Numbers	`DM Mono`	13px	500

Rationale: Serif fonts for reading-heavy question content (higher readability), sans-serif for UI chrome.

9.4 Spacing System

- Base unit: `4px`
- Panel padding: `24px`
- Card padding: `20px`
- Option row height: `52px`
- Palette box: `40×40px` (desktop), `34×34px` (tablet)

9.5 Motion & Animation

- Page transitions: Fade + slight upward slide (200ms ease-out)
- Quote entrance: Fade in over 600ms after hero text
- CTA button: Subtle scale + shadow on hover (transform: scale(1.03))
- Palette color changes: Instant (no transition — mirrors real exam behavior)
- Timer red flash: CSS `@keyframes pulse` at < 5 minutes
- Results page: Score counts up from 0 to final value (1s duration)

9.6 Shadows & Depth

```css
--shadow-card: 0 2px 8px rgba(26, 60, 110, 0.08);
--shadow-panel: 0 4px 16px rgba(26, 60, 110, 0.12);
--shadow-modal: 0 8px 32px rgba(0, 0, 0, 0.2);
```

---

10. Component Library & Interaction Patterns

10.1 Core Components

PaletteBox

```
Props: questionNumber, status (not-visited | not-answered | answered | review |
answered-review)
Size: 40×40px square, 4px radius
Font: DM Mono 13px
Active state (current question): outlined in --color-secondary, bold number
```

OptionRow

TimerDisplay

```
Props: seconds (countdown), totalSeconds
Format: HH:MM:SS using JetBrains Mono
Color: primary when > 10 min, orange 10–5 min, red < 5 min with pulse
```

SectionDropdown

```
Props: sections, selected, onChange
Width: 200px (desktop), full-width (mobile)
Options: "All Sections" + one per section (A, B, C)
Behavior: Filters palette highlight, does NOT jump to a specific question
```

QuoteBlock

```
Container: max-width 640px, centered
Quotation marks: decorative, 60px, color --color-secondary at 30% opacity
Quote text: Playfair Display italic 20px
Attribution: DM Sans 14px color --color-text-secondary
Loading state: skeleton shimmer
```

TestCard (Selection Page)

```
Size: min 200px wide
Contains: Test number, name, question count, status badge, "Start" / "Resume" /
"Review" action
Hover: lift shadow + slight scale
```

10.2 Modal Patterns

- All modals use a semi-transparent overlay (`rgba(0,0,0,0.5)`)
- Modals slide in from bottom on mobile, appear centered on desktop
- Always include a close (×) button

11. Technical Architecture

11.1 Technology Stack

Layer	Technology	Rationale
Framework	React 18 + TypeScript	Type safety, component reuse
Styling	Tailwind CSS 3 + shadcn/ui	Rapid, consistent UI
Build	Vite + Parcel (bundle)	Fast dev + single-file output
PDF Handling	`pdf.js` (Mozilla)	Client-side PDF text extraction
AI API	Google Gemini REST API	User-provided API key
State	React Context + `useReducer`	Session-wide exam state
Persistence	`localStorage`	Caching, progress, API key
Routing	React Router v6	Multi-page navigation
Charts	Recharts	Results visualization

11.2 Folder Structure

```
src/
├── components/
|   ├── exam/
|   |   ├── QuestionPalette.tsx
|   |   ├── QuestionArea.tsx
|   |   ├── TimerDisplay.tsx
|   |   ├── SectionDropdown.tsx
|   |   └── FooterNav.tsx
|   ├── selection/
|   |   ├── TestCard.tsx
|   |   ├── TestGrid.tsx
|   |   └── ProgressDashboard.tsx
|   ├── results/
|   |   ├── ScoreCard.tsx
|   |   ├── QuestionReview.tsx
|   |   └── ExplanationBlock.tsx
|   └── common/
```

```
|       ├── QuoteBlock.tsx
|       ├── Modal.tsx
|       ├── ApiKeyInput.tsx
|       └── Layout.tsx
├── pages/
|   ├── Landing.tsx
|   ├── TestSelection.tsx
|   ├── ExamSimulator.tsx
|   ├── Results.tsx
|   └── Guide.tsx
├── hooks/
|   ├── useTimer.ts
|   ├── useExamState.ts
|   └── useGemini.ts
├── services/
|   ├── geminiService.ts
|   └── pdfService.ts
├── data/
|   ├── examPattern.ts
|   └── fallbackQuotes.ts
└── types/
    └── index.ts
```

---

12. Data Flow & State Management

12.1 Exam Session State (in `useExamState`)

```typescript
interface ExamState {
  testId: string;
  questions: Question[];
  answers: Record<number, string>;   // questionId → selected option
  flagged: Set<number>;              // flagged for review
  visited: Set<number>;              // visited question IDs
  currentQuestion: number;
  timeRemaining: number;             // seconds
  submitted: boolean;
  startedAt: Date;
}
```

12.2 Question Status Derivation

```
Not visited   = id NOT in visited
Not answered  = id in visited, NOT in answers
Answered      = id in answers, NOT in flagged
Review only   = id NOT in answers, id in flagged
Answered+Rev  = id in answers, id in flagged
```

---

13. Performance Requirements

Requirement	Target
Initial page load	< 2 seconds
PDF text extraction (per paper)	< 5 seconds
Gemini quote load	< 3 seconds
Mock test generation (per section)	< 8 seconds
Per-question explanation	< 3 seconds
Timer accuracy	± 1 second
Palette color update	Instant (< 50ms)

---

14. Accessibility & Responsiveness

14.1 Accessibility

- All interactive elements keyboard-navigable (Tab + Enter)
- ARIA labels on palette boxes and option rows
- Color is never the ONLY indicator (always paired with text or icon)
- Timer announced by screen reader at 10-min and 5-min mark
- Minimum contrast ratio: 4.5:1 for body text

14.2 Responsive Breakpoints

Breakpoint	Layout Change
`≥ 1280px`	Full 3-panel layout as spec'd
`768–1279px`	Palette collapses to overlay/drawer
`< 768px`	Palette as bottom sheet; question full-width

---

15. Out of Scope (v1.0)

- User authentication / login system
- Backend database or server
- Multi-user leaderboards
- Offline Progressive Web App (PWA) mode
- Push notifications
- Admin CMS to manage PDFs
- Payment or subscription gating

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16. Open Questions

#	Question	Owner	Priority
1	What are the exact 21 PDF filenames?	User	P0 — Blocks paper naming
2	How is the APICET large PDF divided by year?	User	P0 — Blocks Set 1/2 split
3	Should Grand Mock tests regenerate each time or only once?	User	P1
4	What is the student's name to pre-fill in the exam header?	User	P2
5	Should the portal support multiple user profiles locally?	User	P2
6	Are there any specific subjects from particular PDFs to emphasize in mocks?	User	P2

#	Question	Owner	Priority
7	Should there be a dark mode toggle?	User	P3

Appendix A — Reference UI Analysis

The provided reference screenshot (exam interface from an existing portal) confirms:

- **Left panel:** Question palette with numbered grid (1–60 visible), color-coded status
- **Top center:** Section dropdown (e.g., "ARITHMETIC"), user name displayed
- **Top right:** Timer in HH:MM:SS format (orange color)
- **Question area:** Clean white, question text in readable font, 4 radio options as full-row selectors
- **Bottom bar:** Prev (amber), Clear (grey), Submit (green), Review, Next (orange) — evenly distributed
- **Legend at bottom-left:** Color explanation for Not Visited, Not Answered, Answered, Answered Review

This portal will replicate and enhance this pattern with higher visual polish and AI-powered features.

End of PRD v1.0 — ICET 2026 Mock Preparation Portal